
Supplementary information

Hallmarks of transcriptional intratumour heterogeneity across a thousand tumours

In the format provided by the
authors and unedited

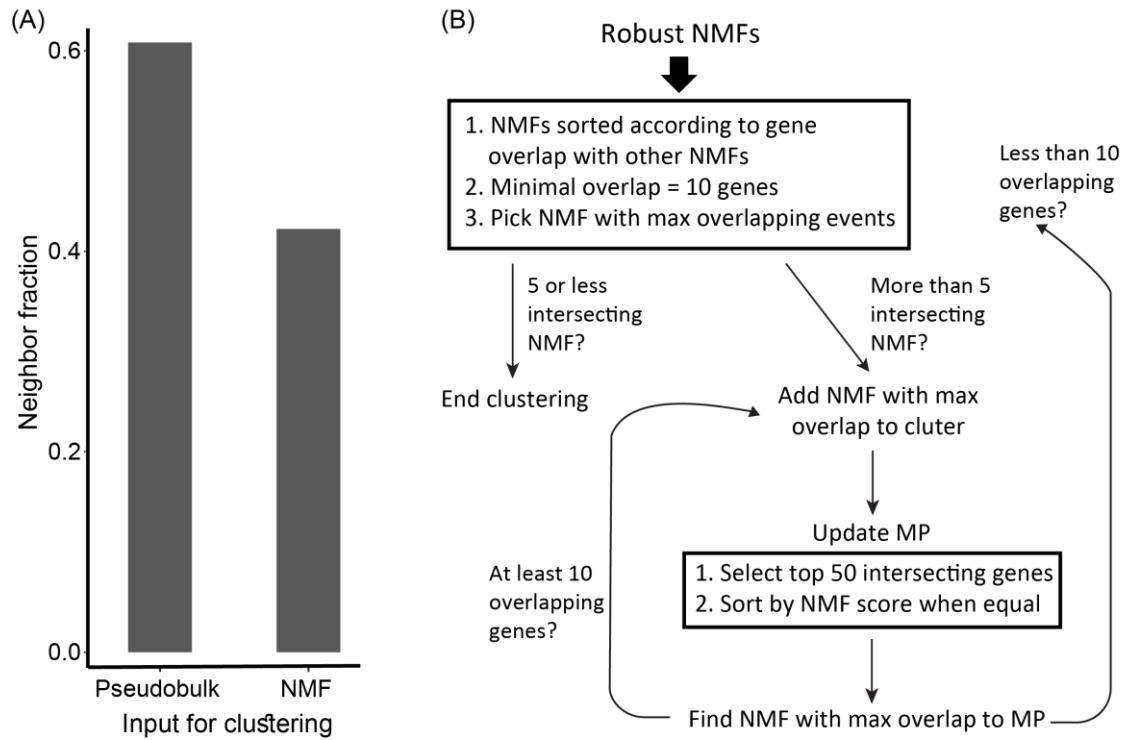


Figure S1. Clustering robust NMF programs. (A) Comparison between the fractions of neighboring programs (i.e. immediately adjacent in the hierarchical clustering dendrogram tree) that originate from the same study, in the NMF vs. in a pseudobulk analysis. The fraction in the NMF approach is significantly lower ($P=1.84e^{-31}$), indicating a reduced bias to identify similarities within the same study (i.e. the same batch). (B) Diagram describing the clustering process for robust NMFs. 5547 robust NMF programs were selected for initial clustering, out of which 3163 programs formed clusters and participated in the meta-programs (Table S2). See methods for further description.

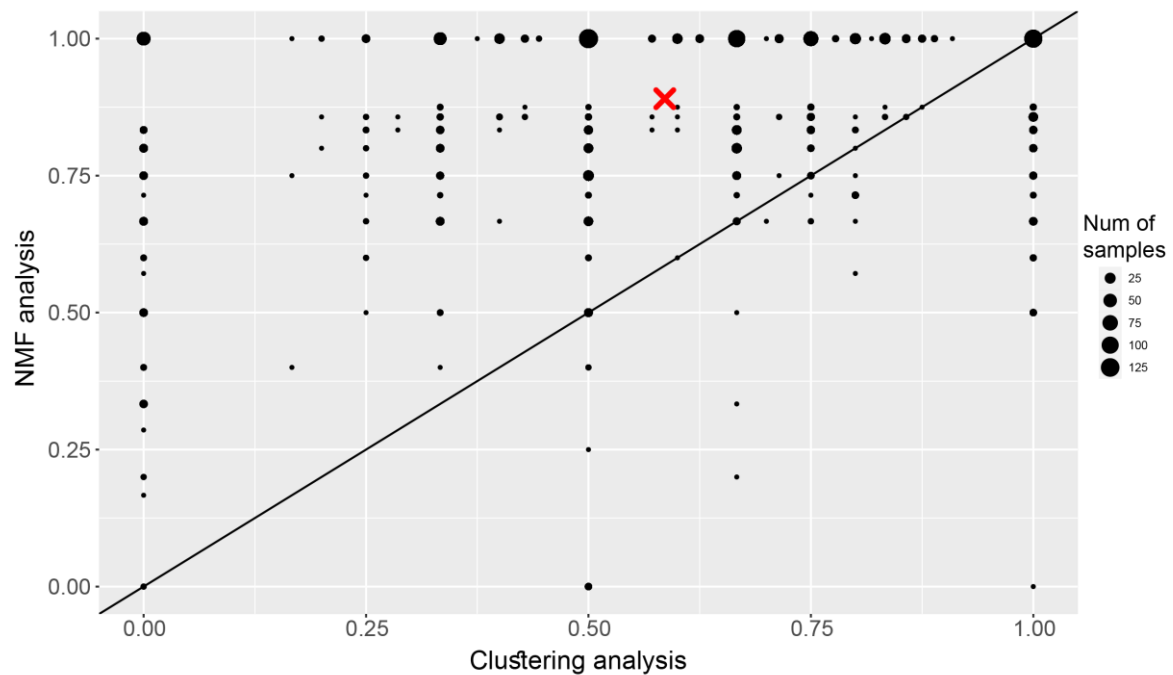


Figure S2. Fraction of ITH programs (defined by NMF or by Louvain clustering) that may be accounted for by the meta-programs. The Y-axis shows, for each sample, the percentage of robust NMF programs identified in that sample that have a significant positive score for at least one MP. The X-axis shows a similar analysis with an alternative method to define ITH programs, based on Louvain clustering (with K=10). Cells in each Louvain cluster were scored against the MPs, and the fractions of clusters from each sample where more than 50% of the cells scored positive was computed. Since many samples had the exact same X,Y values, the number of samples in each coordinate is reflected by the size of the circle, as noted in the legend at the right.

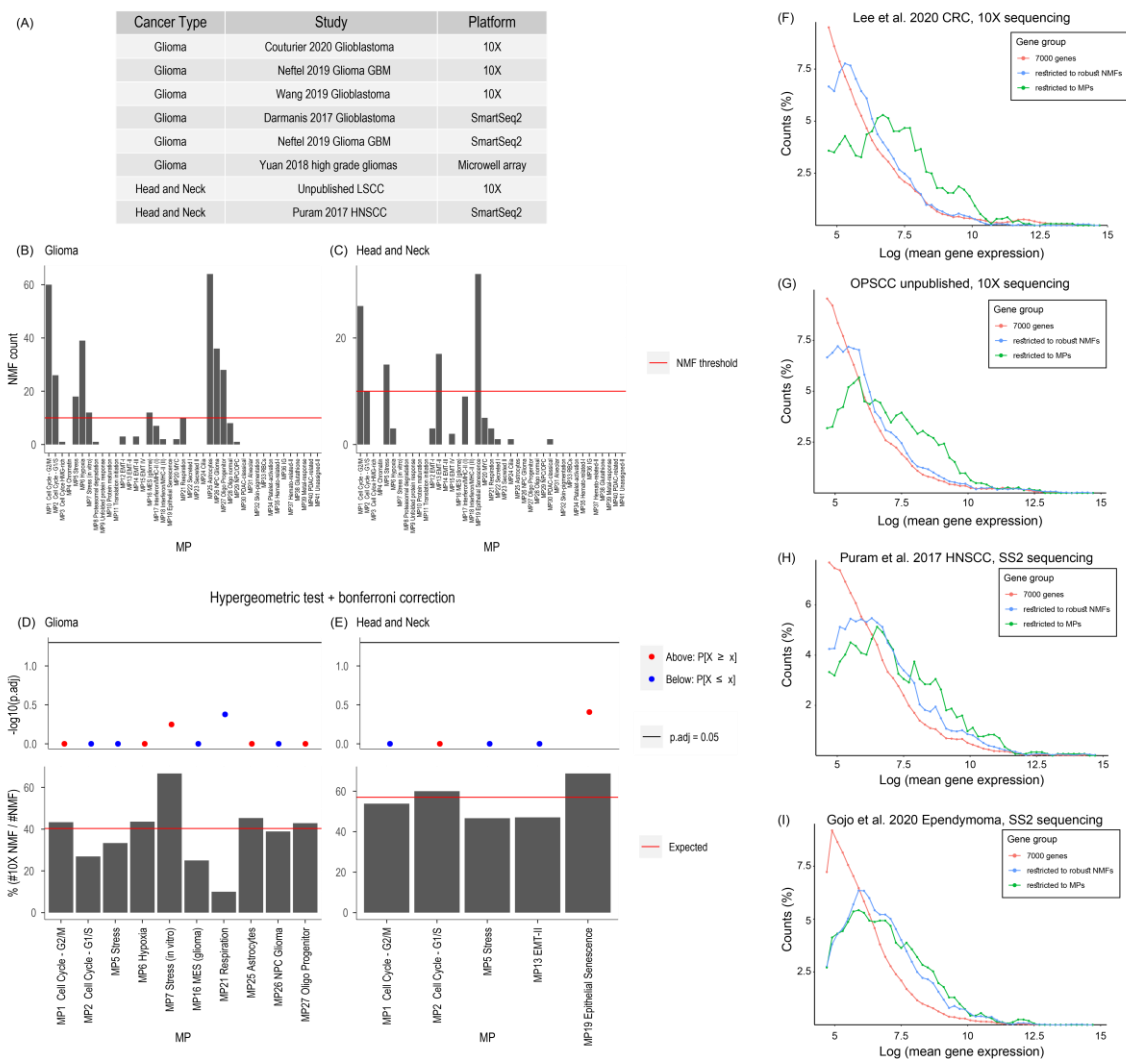


Figure S3. Technical controls for potential biases by platform and expression level. (A) Table summarizing the selected studies for this analysis. **(B-C)** Number of NMF programs derived from glioma and HNSCC samples for each MP. MPs with at least 10 NMF programs were selected (above the red horizontal line). **(D-E)** The expected percentage of NMF programs from samples sequenced by 10X is delineated by the red horizontal lines. Upper panels show the significance (red above the expected and blue below). None of the cases were statistically significant. **(F-I)** Distributions of average expression levels for the top 7000 genes (on which NMF was performed) and for genes selected for NMF programs and MPs. Upper 2 studies were sequenced with 10X, and lower 2 panels with Smart-seq2.

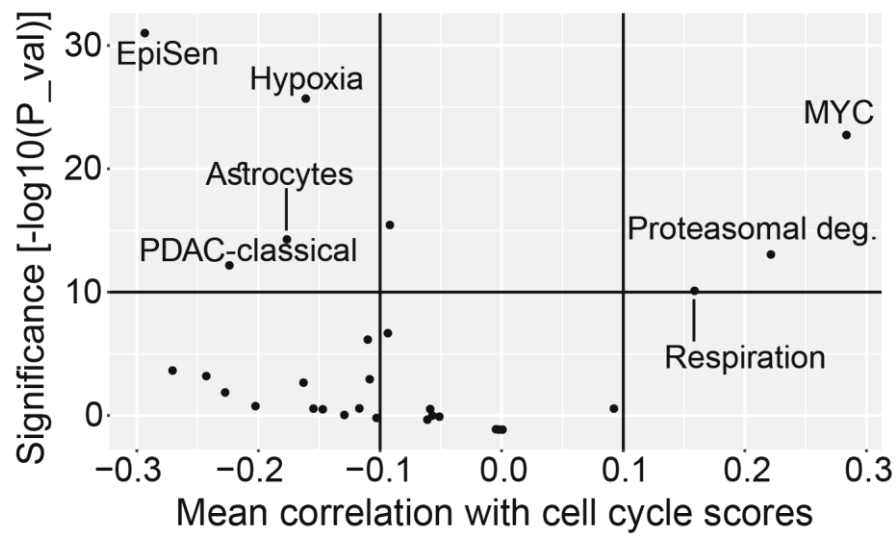


Figure S4. Meta-program associations with proliferation. For each tumor, the correlation between cell scores for the cell cycle MPs (maximum over the scores for MP1-4) and to different MPs was calculated. The mean values of these correlations across all tumors are shown (X-axis). Significance was then defined for each MP by a one-sided t-test (Y-axis), adjusted by Bonferroni correction. The horizontal and vertical lines delineate the thresholds above (or below) which we considered values to be highly significant.

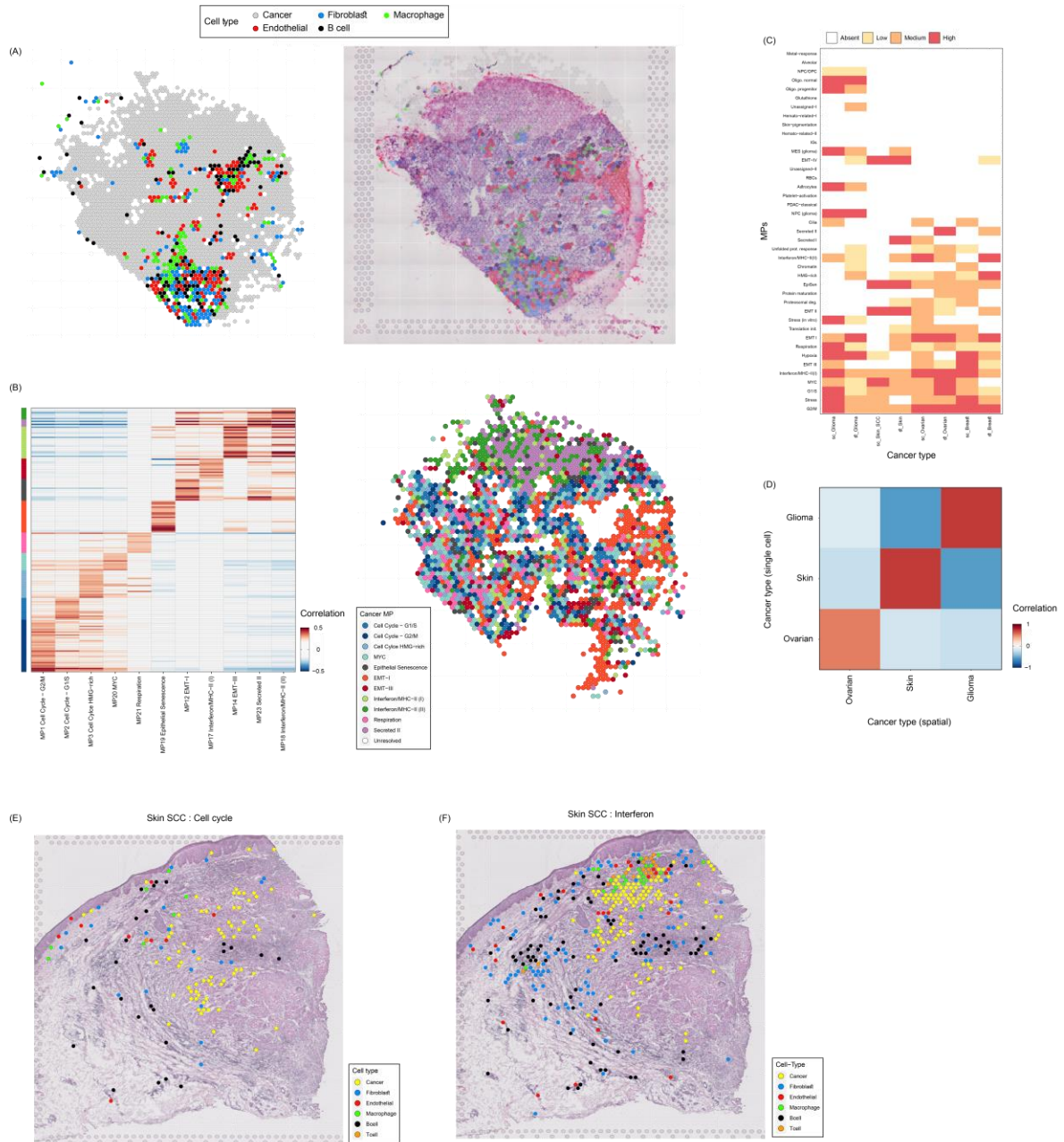
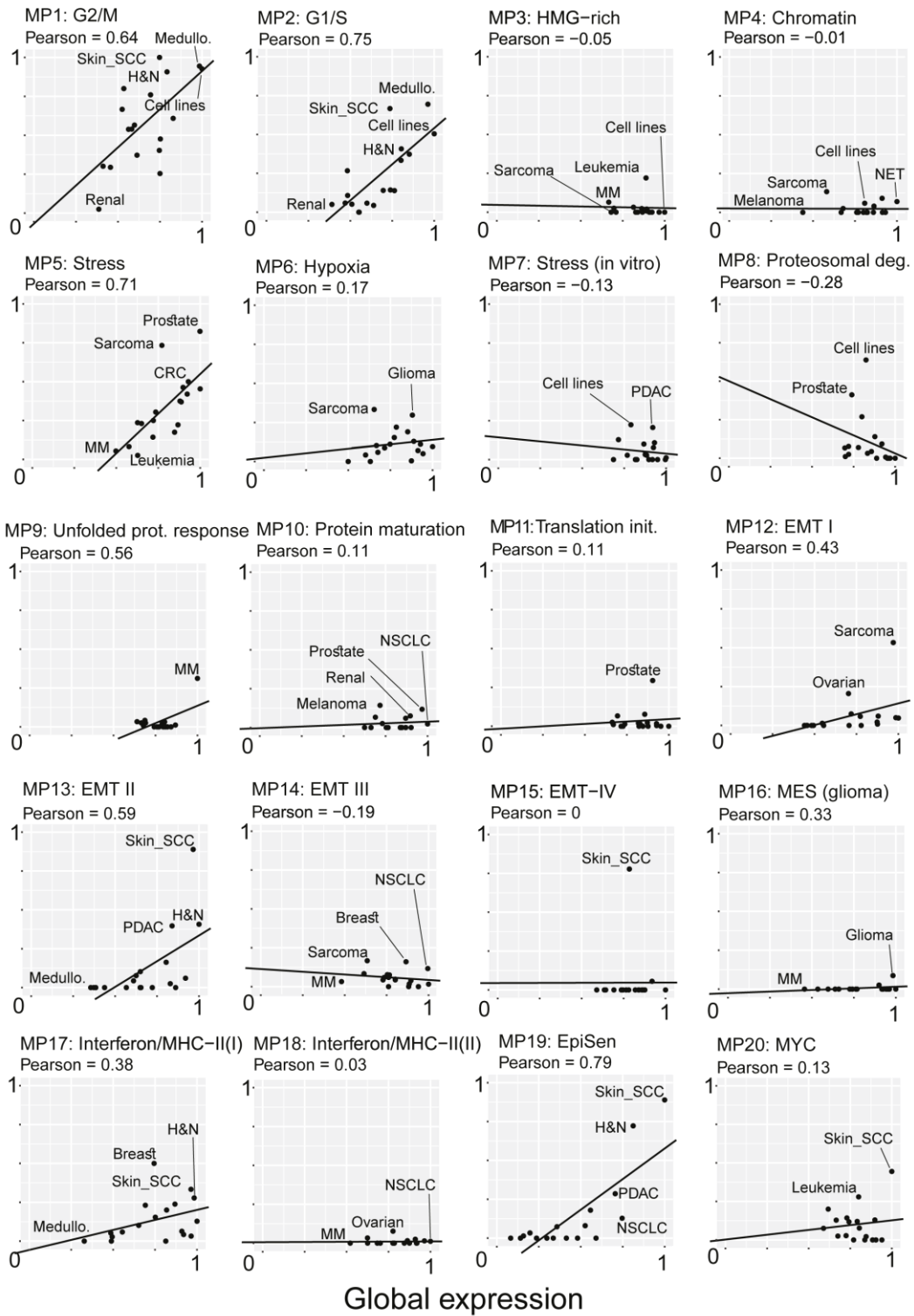
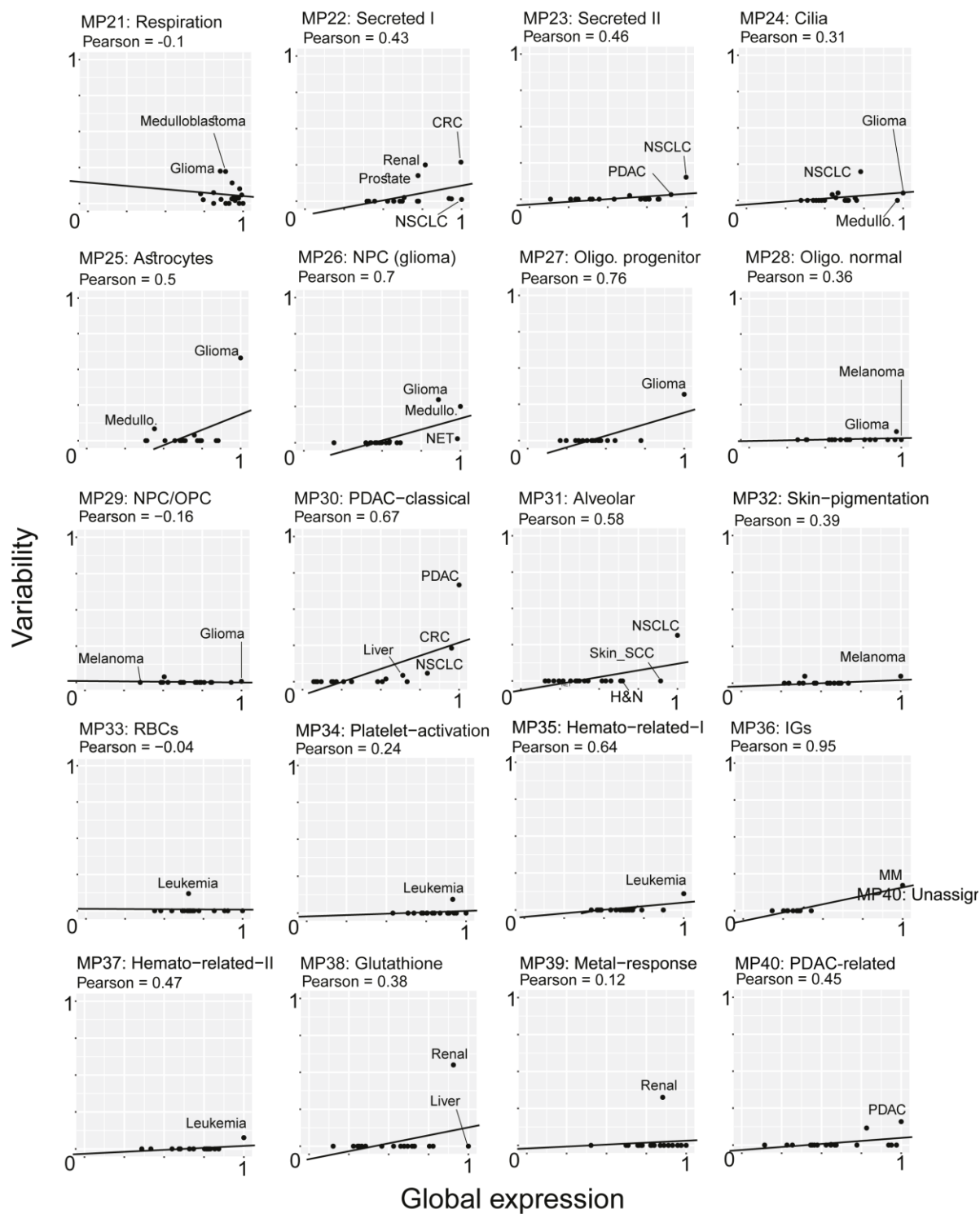


Figure S5. Analysis of MP expression in spatial transcriptomics data. (A) Visium image of an ovarian sample showing the spatial organization of spots assigned to different cell types. **(B)** Correlation between overall MP scores and specific MP genes (rows), for MPs (columns) with at least 10 genes with correlation above 0.3. Correlations were correlated across spots of the ovarian cancer sample shown in A; right panel shows the same sample, with assignment of malignant spot to MPs based on maximal score. **(C)** Abundance of MPs in scRNAseq data (denoted as 'sc') and in spatial transcriptomics data (denoted as 'st'), for glioma, skin and ovarian cancer. Abundance categories are defined as in **Fig. 3A**. **(D)** Correlations between the single-cell and the spatial-transcriptomics columns in (C), demonstrating consistency in the identification of MPs in each of the three cancer types. **(E-F)** Co-localization of spots assigned to the cell cycle (left) and Interferon response (right) MPs, for malignant and TME spots in a selected skin sample.

Variability





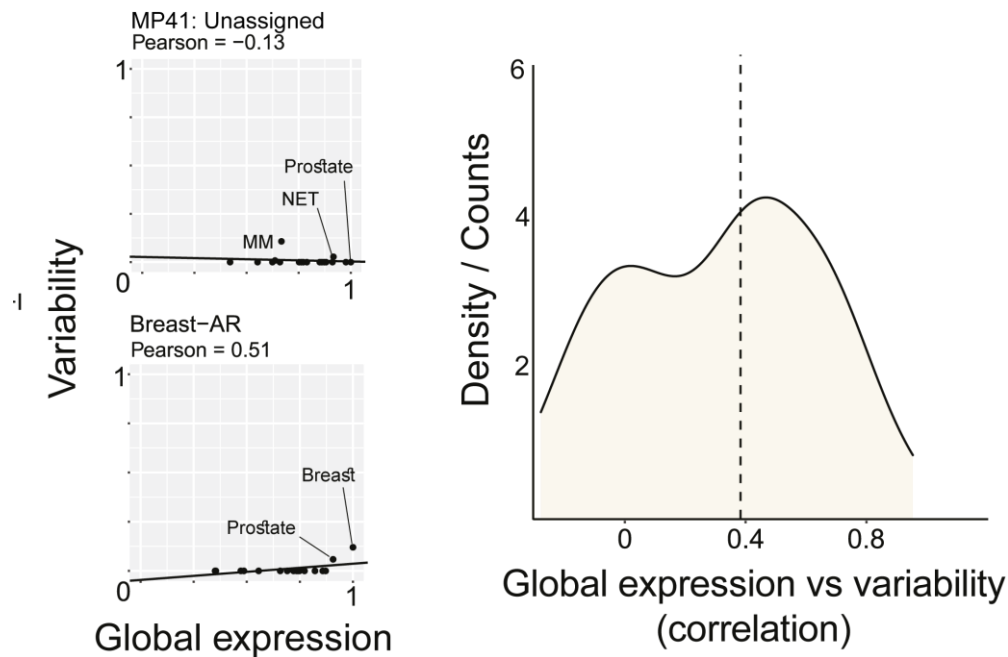


Figure S6. Global (mean) expression vs. variability of MP expression across different cancer types. Each panel shows global expression (X-axis) and variability of expression (Y-axis) of a specific MP across all cancer types analyzed (dots), along with a Pearson correlation and labels for selected cancer types. Global expression is defined as the mean expression of the MP genes across malignant cells from all samples of a particular cancer type; global expression is further normalized into [0..1]. Variability of the MP in a given cancer type is defined as the fraction of tumors of that cancer type in which at least one of the robust NMF programs was assigned to the MP. The last panel shows the distribution of correlations between global expression and variability across all MPs, with a dashed line indicating the median correlation ($R=0.38$). See methods for further description. This analysis reveals that on average there is a correlation between global expression and variability but also highlights many exceptions, including the following:

- (i) Interferon response is particularly variable (but moderately expressed) within breast cancer.
- (ii) Respiration and hypoxia are both particularly variable in glioma, but has intermediate expression in these contexts.
- (iii) Proteasomal degradation and translation initiation are most variable in prostate cancer as well as in cell lines, despite lower than average expression in these contexts.
- (iv) MYC-targets, which is most often variable in leukemia but is higher in many other contexts. Notably, MYC targets have both the highest activity and the least variability in cell lines, perhaps indicating that MYC activity is selected for during the establishment of cell lines such that its variability is avoided in culture.

EpiSen

HNSCC

SPRR1B p63

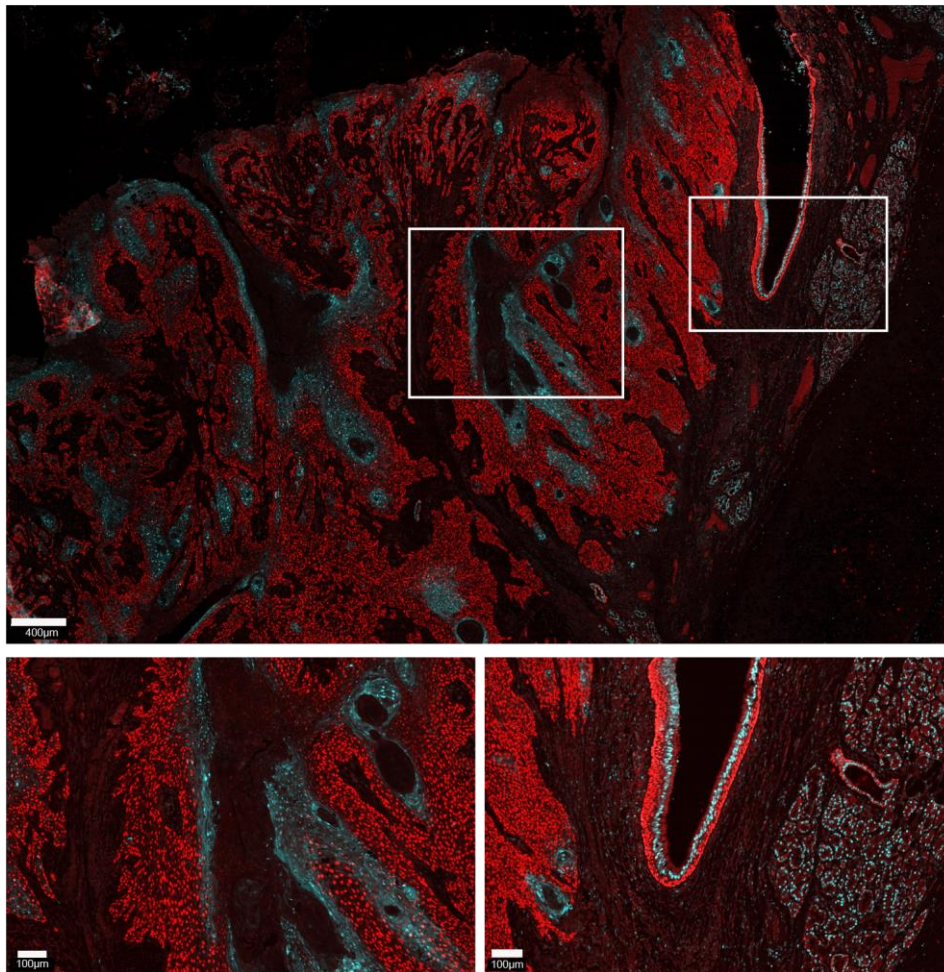


Figure S7. EpiSen represents an aberrant form of normal epithelial differentiation. Right lower panel shows layers of normal differentiating epithelium with an organized layer of SPRR1B+ cells adjacent to p63+ cells. Left lower panel reveals a disorganized malignant version of a similar structure (representative images from 3 independent experiments).

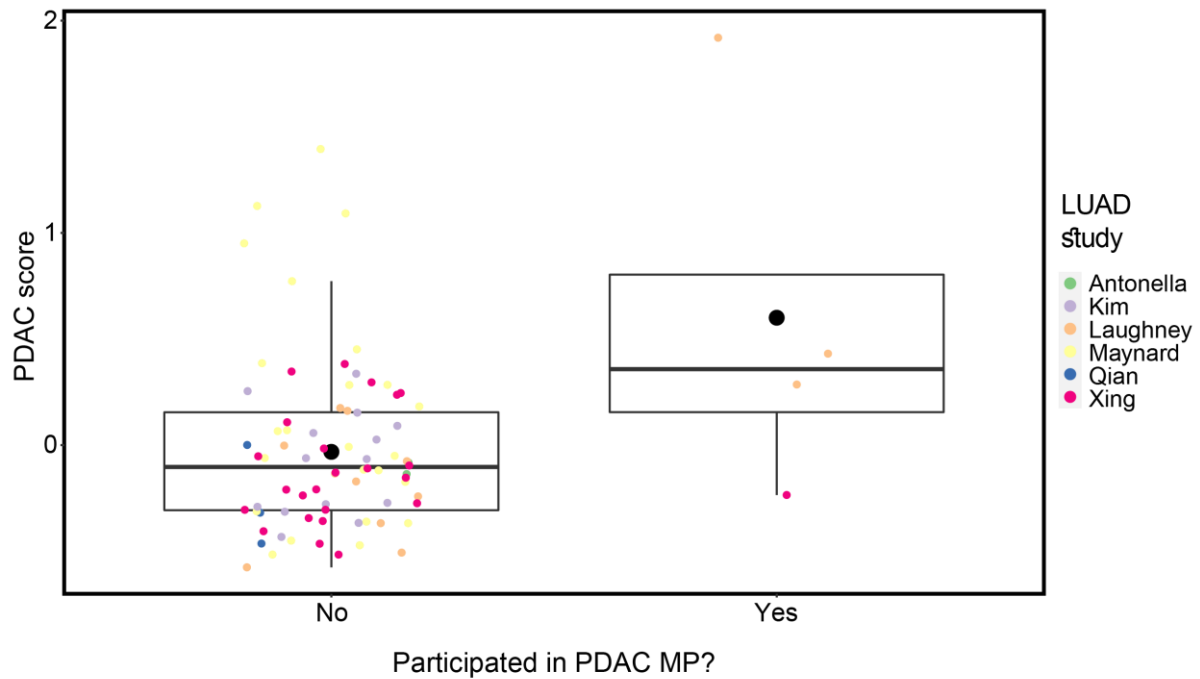


Figure S8. Overall expression of PDAC genes in lung adenocarcinoma (LUAD) samples. The top 100 genes by higher expression in PDAC than in LUAD samples were defined as a PDAC signature. Average expression of the PDAC signature genes in malignant cells from each tumor was defined as a PDAC score. The PDAC scores were centered and are shown by box plots for two subsets of LUAD samples (n=80 samples in total), based on whether MP30 (PDAC-classical) was identified as variable within these tumors. Boxes indicate the median and 1st and 3rd quartiles, while the upper and lower whiskers respectively extend to maximal and minimal values which are no further than 1.5 times the interquartile range from the 3rd and 1st quartiles. LUAD Samples with MP30 variability had significantly higher PDAC scores ($P=0.002$, one-sided t-test). Sample are also colored by their study as indicated at the right legend.

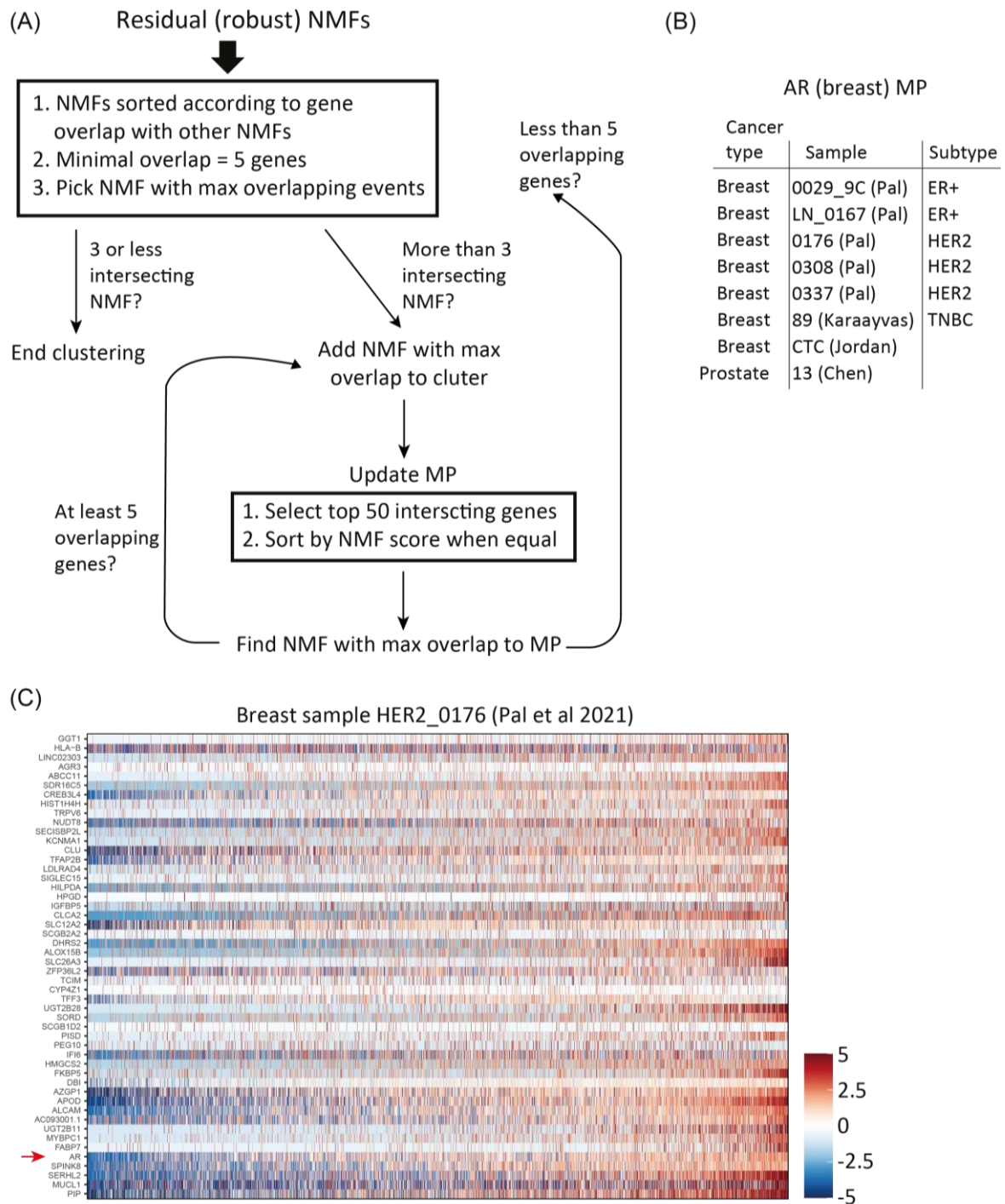


Figure S9. Generation of the Lenient-defined meta-programs. (A) Similar to the process described in figure S1A, the robust NMFs that were not included in the initial MPs were clustered using more lenient parameters. 59 additional MPs were thus generated (**Table S2**). (B) List of samples in which MP50 was identified. This MP was identified mostly in breast cancer samples, it is highly enriched for the Androgen Response (AR) Hallmark signature (**Table S3**) and contains the gene encoding AR. (C) Heatmap for the HER2_0176 breast cancer sample from Pal et al. (2021), showing relative expression of MP50 genes (rows) across malignant cells (columns) ordered by MP50 expression. The AR gene is highlighted by a red arrow.

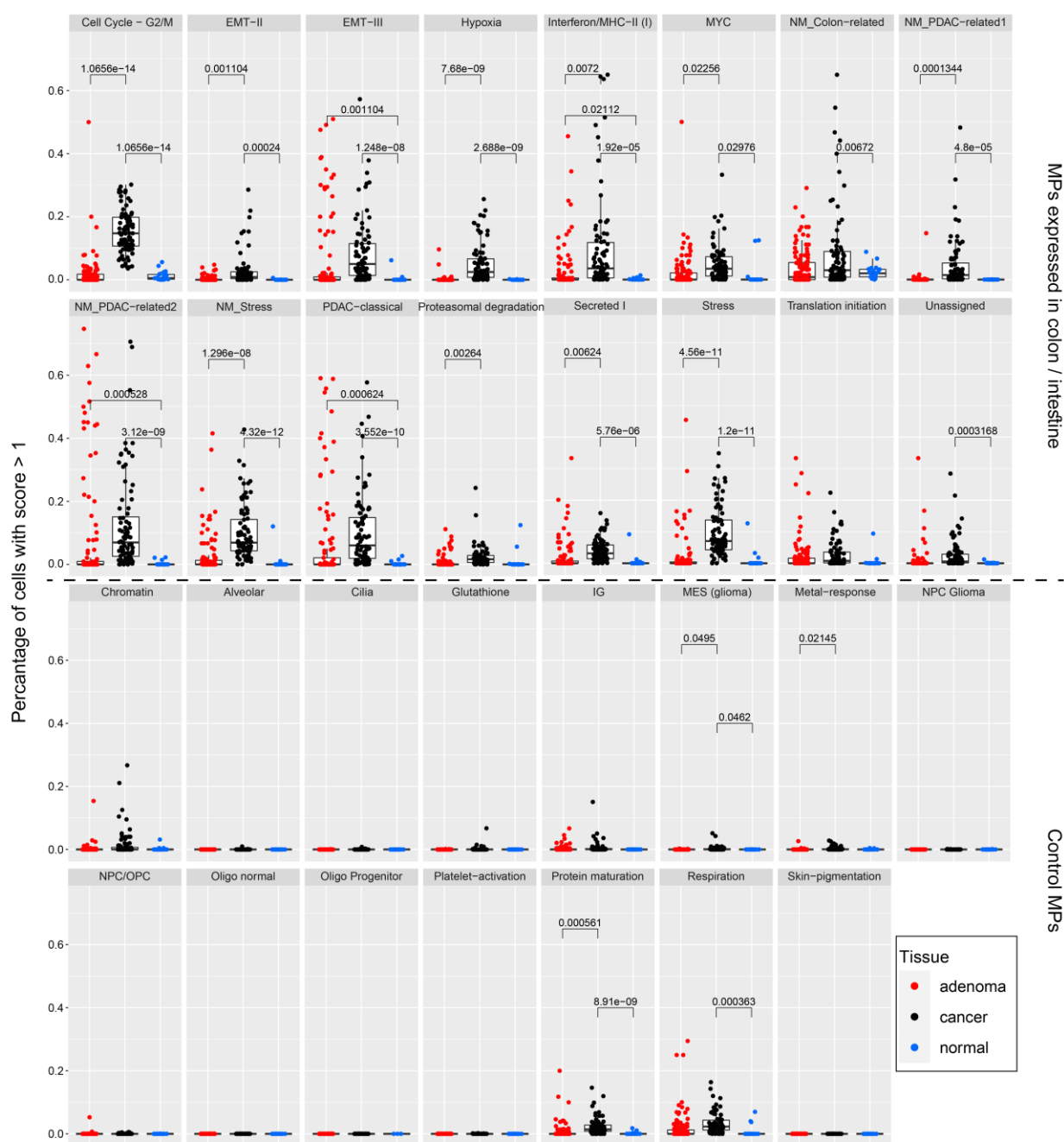


Figure S10. MP expression in pre-cancerous tissue. For each MP (panel), box plots show the percentages of cells that scored above 1 in pre-malignant (red, n=76 samples), malignant (black, n=84 samples) and normal (blue, n=21 samples) tissue samples. MPs above the dashed line are those detected in colon and those below the lines were not detected in colon and are added as a control for technical differences between tissue types. ‘NM’ pasted at the beginning of some MP names stands for non-malignant. P-values were calculated by a two-sided t-test with Bonferroni correction. Only significant cases are highlighted by showing the p-value. Boxes indicate the median and 1st and 3rd quartiles, while the upper and lower whiskers respectively extend to maximal and minimal values which are no further than 1.5 times the interquartile range from the 3rd and 1st quartiles.

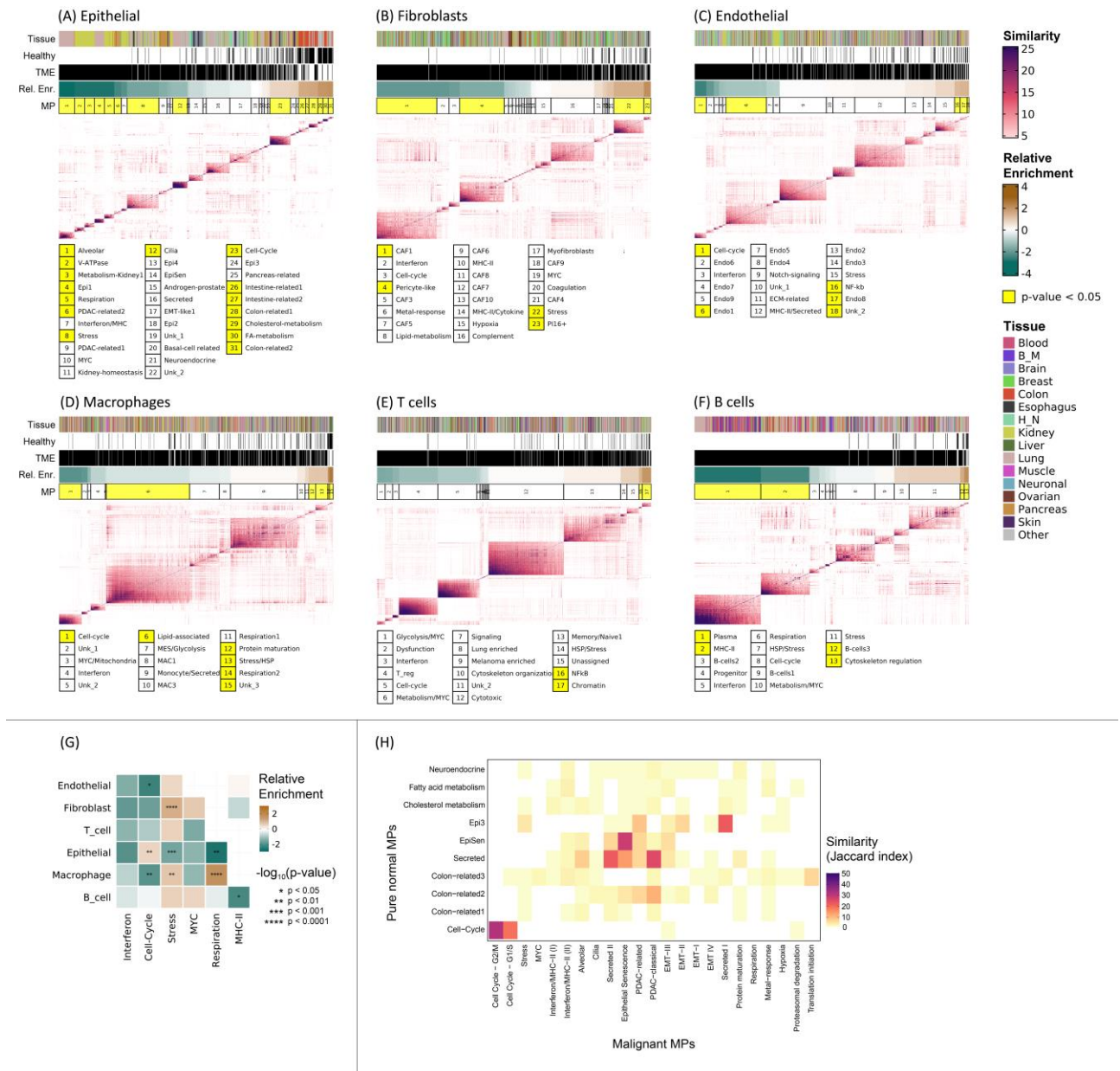


Figure S11. Meta-programs in normal (healthy) vs. tumor tissues. (A-F) Each panel shows the clustering of NMF programs to define MPs in one cell type. NMF programs were derived from both normal and malignant datasets. Heatmaps show their similarities (by Jaccard index) and the clustering that defines MPs (with MP borders shown in the panel above). Top panel also shows the origin of each NMF program – the tissue type and whether it is from healthy or tumor (TME) sample, and the relative enrichment of each cluster (i.e. MP) in healthy vs. tumor samples. Enrichment was calculated as $\log_2(\text{observed \# NMFs} / \text{expected \# NMFs})$ and corresponding P-values were calculated by hyper-geometric test and corrected for multiple comparisons using FDR. Significant enrichments are highlighted in yellow. MPs and their annotations are listed below each heatmap. (G) Relative enrichments in normal vs. tumor tissues of the MPs that are observed across different cell types. (H) Overlap between malignant MP clusters and MP clusters derived using solely normal NMFs.

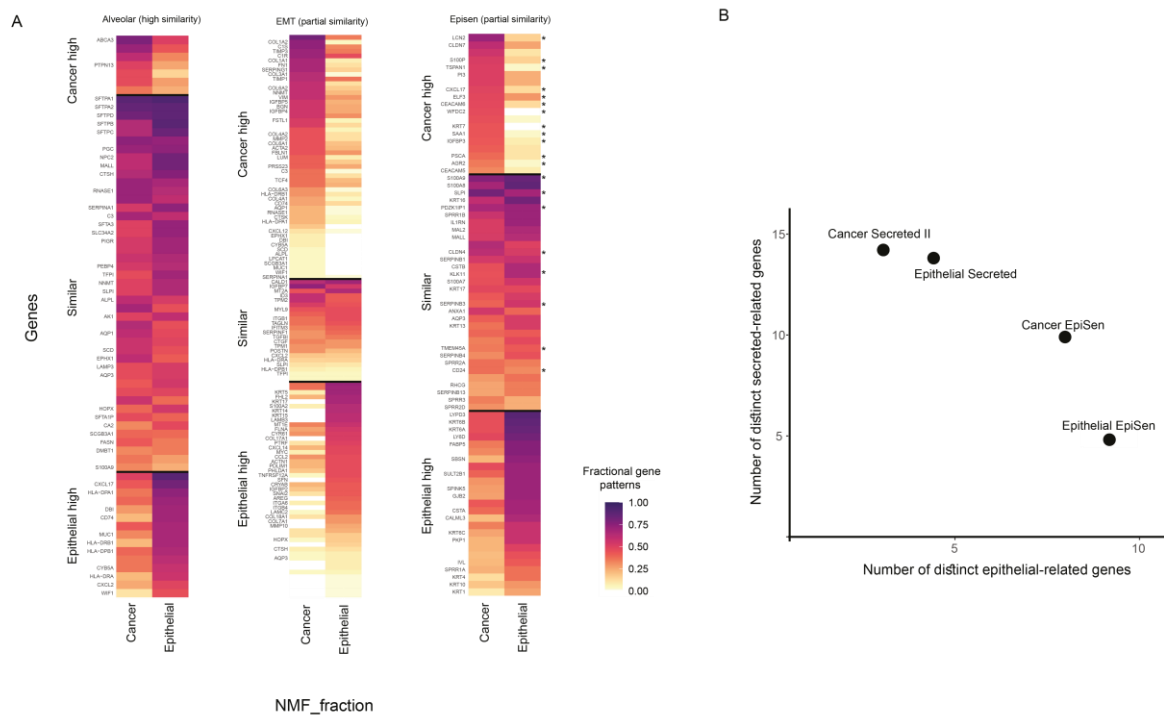


Figure S12. Comparison between related MPs from malignant and non-malignant epithelial cells. (A) Each panel shows a comparison of one pair of related MPs, from carcinoma (left column) and non-malignant epithelial cells (right column), including all genes (rows) included in either of the MPs. Heatmap colors reflect the fraction of NMFs from a given MP in which the respective gene has an NMF score above a given threshold (see Methods for more details). Black horizontal lines separate gene groups according to their differential expression patterns (i.e. higher in malignant NMFs, higher in epithelial NMFs or similar in both). In the EpiSen (right) panel, asterisks indicate genes that are also included in the secreted MP. **(B)** The EpiSen MP of cancer cells contains both epithelial-related and secreted genes (see gene lists below), while these two groups of genes are largely represented by separate MPs in epithelial non-cancer cells. Scatter plot shows the number of epithelial-related genes (X-axis) and number of secreted genes (Y-axis) in four relevant MPs. For each MP, the number of genes in each class was defined as the number of genes with non-zero NMF scores, averaged across the constituent NMF programs of that MP.

Epithelial-related genes: KRT1,KRT4,KRT10,KRTDAP,KRT6C,KRT6A,KRT6B, CSTA,PKP1, IVL,CALML5,DMKN,KRT19

Secreted-related genes: CXCL2,CXCL1,CXCL17,CXCL8,SERPINF1,MUC5B,CLU, WFDC2,C3,BPIFB1,MDK,SCGB1A1,MGP,TNFSF10,TFF3,SAA1,MSMB,LTF

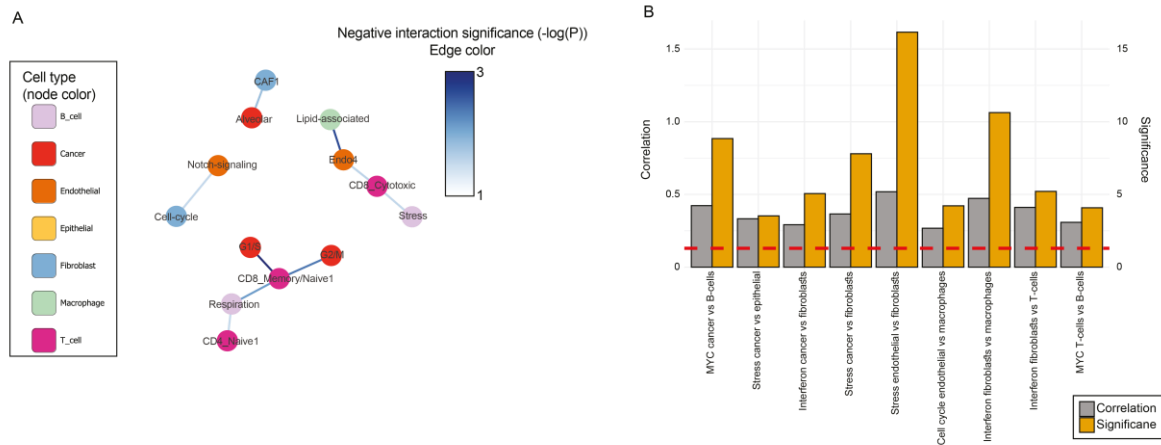
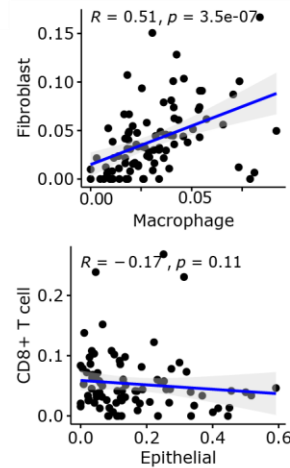
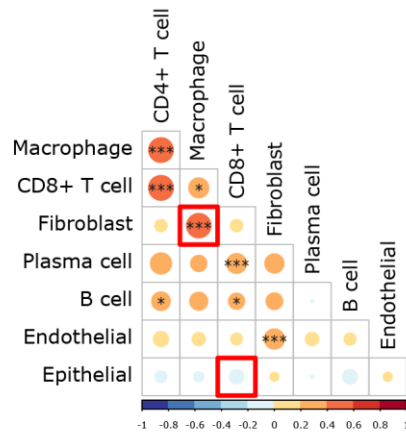
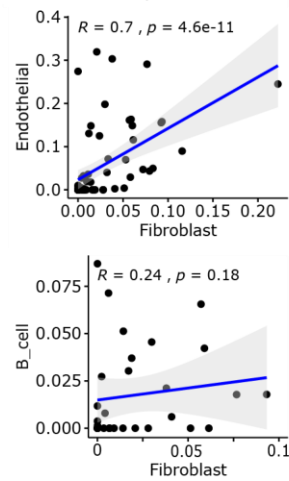
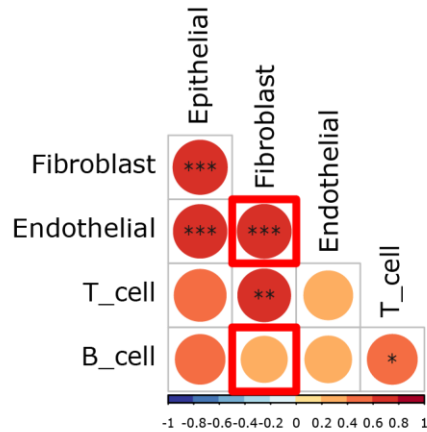


Figure S13. Association of malignant and TME MPs. (A) Similar to Fig. 5A, showing interactons with significant negative correlations between MPs of different cell types (B) Significant correlations between MPs reflecting similar functions, in different cell types, are maintained after removing overlapping genes. We repeated the correlation analysis shown in Fig. 5A and found that in almost all cases, the interactions remained significant even after removing overlapping genes (up to 40 out of the 50 MP genes). Shown are selected cases. Grey bars represent the new correlations (after removing overlapping genes) whose values are shown in the left vertical axis, and yellow bars represent the new significance values ($-\log_{10}(\text{p-value})$) whose values are given in the right vertical axis. Red broken line demarcates the significance value that corresponds to $p\text{-value} = 0.05$. P-values were calculated based on Pearson correlation coefficient and adjusted by Benjamini-Hochberg correction.

Schuerch et al.



Zhang et al.



Blise et al.

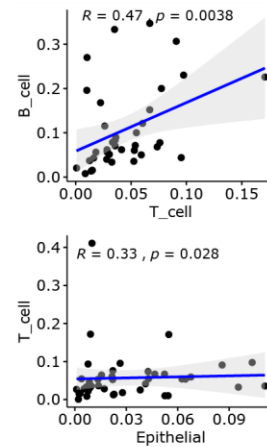
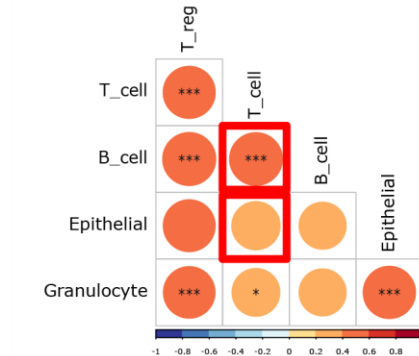


Figure S14. Proximity of cycling cells from different types observed with CODEX data.

Left panels show quantification of the correlation between cycling cells of different types, across three CODEX studies. Color denote Pearson correlation and asterisks denote its significance (1-3 asterisks reflect $P < 0.05$, 0.01 and 0.001 respectively) assessed using a two-sided t-test. Representative examples are marked by red squares and shown as scatter plots in the right panels. The error bands depict the 95% confidence interval around the mean.

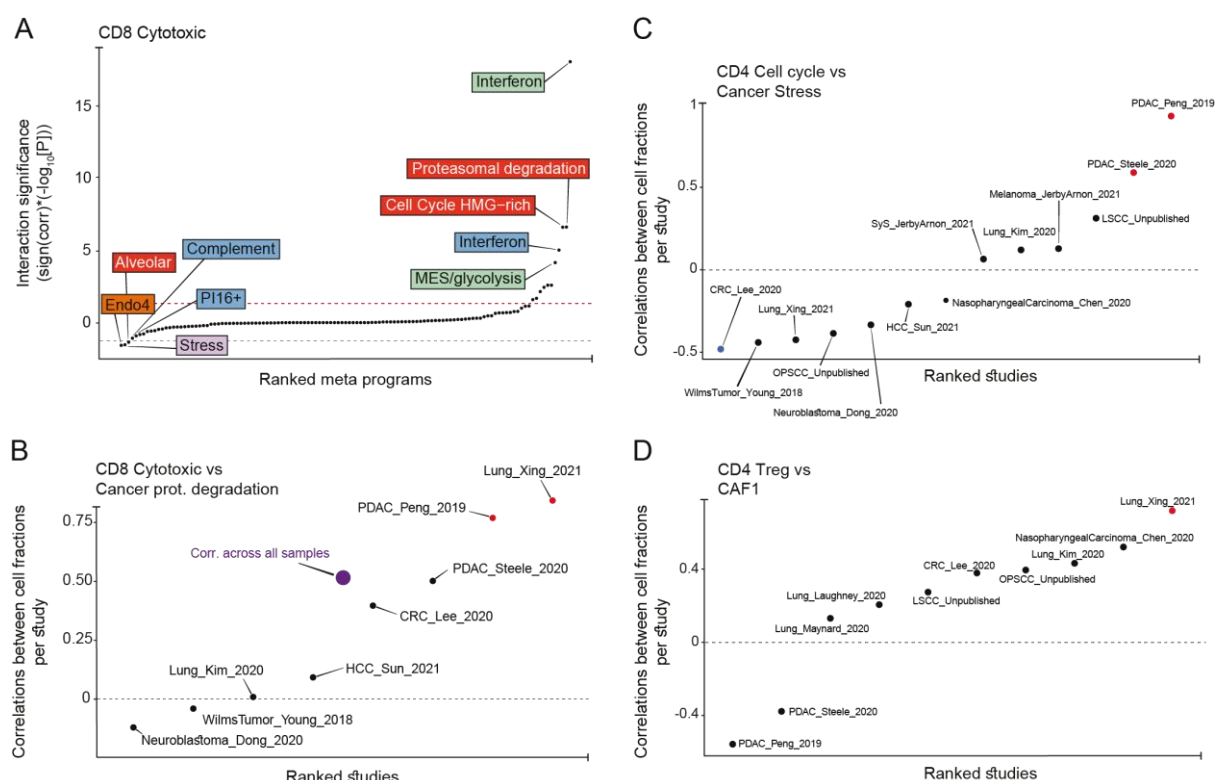


Figure S15. Variability of the MP interactions. **(A)** Significance of Pearson correlations in the integrated dataset (Y-axis) is shown for all MPs (ranked on X-axis) with CD8 cytotoxicity. P-values were adjusted by Benjamini-Hochberg correction. **(B)** Correlations (Y-axis) within individual studies (ranked on X-axis) for the interaction between CD8+ Cytotoxicity MP and malignant cell proteasome degradation MP. The correlation across all samples is highlighted in purple. **(C-D)** Same as (B) for the interaction between the CD4+ Cell cycle MP and the malignant stress MP and for the interaction between the CD4+ Tregs MP and the CAF1 MP. Red and blue dots respectively reflect statistically significant positive and negative correlations ($P < 0.05$), based on Pearson correlation coefficient (an asymptotic confidence interval is given based on Fisher's Z transform).

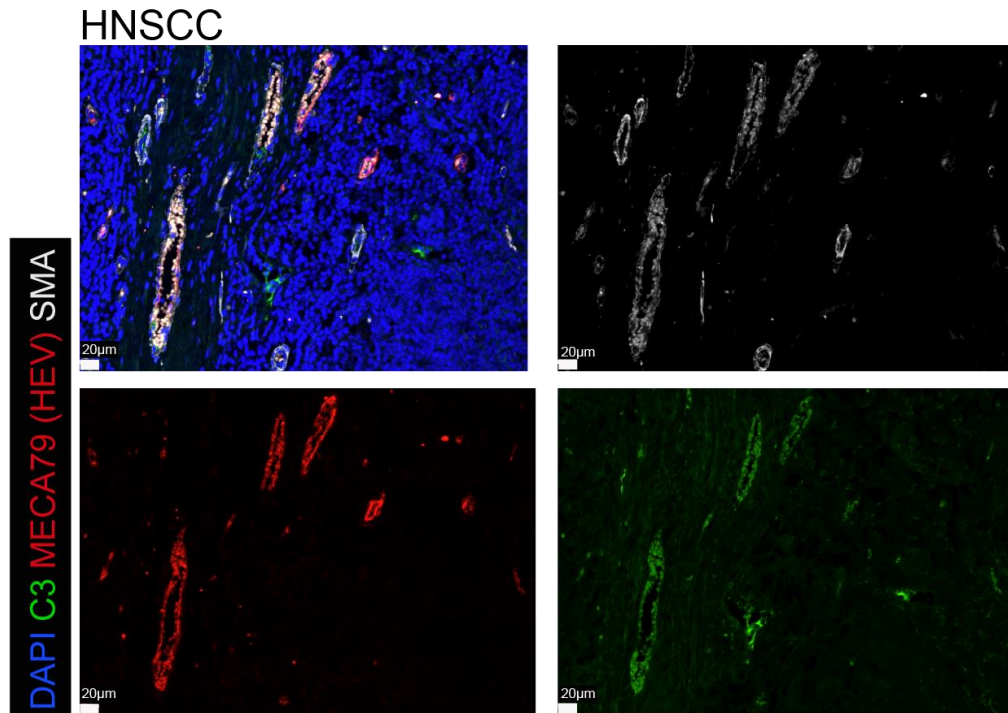


Figure S16. Experimental validation of interaction between the fibroblast complement MP and the endothelial HEV MP. High endothelial venules (MECA79+) are surrounded by perivascular fibroblasts (SMA+) with upregulated expression of complement (C3+) (representative images from 3 independent experiments).